

Below is a list of BBPV (Boiler and Pressure Vessel) Codes organized by category – now including requirements, responsible compliance officials, and relevant standards.

1. Certification & Authorization

- ✓ ASME Stamp (U, S, H) is mandatory for all new pressure vessels and boilers.
- ⚠ Unauthorized fabrication or repair without ASME/NBIC inspection is a code violation and often illegal.
- ✓ National Board R Stamp is required for any repair to a pressure-retaining item.
- ✓ Licensed Operators (Red Seal, State BOE, etc.) must operate high-pressure systems in most jurisdictions.

3. Combustion & Burner Safety

- ✓ Pre-purge, ignition sequence, and flame safeguard interlocks are mandatory per NFPA 85 and CSD-1.
- ⚠ Bypassing flame scanners, safeties, or using manual overrides without lockout protocol is a major violation and can result in explosions.
- ✓ All combustion enclosures must be sealed, flame-resistant, and interlocked to stop fuel during access.

4. Installation & Piping

- ✓ Must meet ASME Section I or VIII + NBIC Part 1 for all piping and support systems.
- ⚠ No black iron gas piping inside buildings unless code-approved and sealed.
- ✓ Relief valves must be certified, set to MAWP, and sized to prevent overpressure.
-

5. Safety Devices & Controls

- ✓ Pressure relief valves must be tested annually and tagged with date & tech name.
- ✓ Emergency shutdown switches must be accessible, labeled, and tested monthly.
- ⚠ Operating a boiler without active low-water cutoffs (LWCO) or high-limit switches is grounds for immediate shutdown.

2. Material Testing & Traceability

- ✓ All pressure parts must have traceable Mill Test Reports (MTRs).
- ⚠ Welding rods and filler metals must match P-Numbers and WPS specs – no substitutions.
- ✓ Hydrostatic testing must be done at 1.5x MAWP before commissioning.
- ⚠ Failure to perform or record NDT (RT, UT, PT, MT) before startup invalidates ASME code compliance.

6. Documentation & Records

- ✓ Daily logs for pressure, temp, flame signal, and feedwater levels are required by most inspectors.
- ✓ WPS (Welding Procedure Specs), PQR (Procedure Qualification Records), and WPQ (Welder Performance Qualification) must be on file for all shop or field work.
- ⚠ Missing documentation or falsified records can trigger insurance denial and OSHA fines.

⚠ 7. Environmental & Regulatory

- ✓ Must follow OSHA 1910 and EPA Clean Air Act for emissions if above thresholds.
- ✓ SDS sheets for water treatment chemicals, fuels, and degreasers must be posted and available on site.
- ⚠ Discharging boiler blowdown or chemicals into storm drains or unapproved lines is illegal under EPA guidelines.

8. Emergency & Inspection Must-Haves

- ✓ Annual inspections by Authorized Inspector (AI) or State Boiler Inspector are required.
- ✓ Confined Space Entry Permit & Hot Work Permit must be completed for any work inside the drum, furnace, or tank.
- ⚠ Blocked safety valves, welded shut manways, or missing nameplates are immediate red flags and may result in system lockout.